

Practice MCQs for Nutritional aspects of pregnancy, lactation and infant nutrition with explanations.

1. A 21-year-old woman comes to the physician because of a 3-week history of missing her period. Laboratory studies confirm that she is pregnant. She says that she is still drinking red wine as she believes that it will not harm her fetus. The physician recommends not to drink any alcohol as this can lead to physical, cognitive and behavioral disabilities in her child. Which of the following is most frequently a complication if she continues to take alcohol?

- A. Protruding tongue
- B. Macrosomia
- C. Facial abnormalities
- D. Cleft lip
- E. Spina bifida

2. A researcher is studying composition of human milk and observes that it contains the proteins, whey and casein. Together they provide all dietary essential amino acids and in addition amino acids that are especially needed in large amounts in infants for collagen synthesis. Which of the following amino acids found in human milk is most likely used for this function?

- A. Proline.
- B. Leucine.
- C. Cysteine.
- D. Tyrosine.
- E. Histidine.

3. A boy is delivered at full-term in the hospital. He is healthy and starts breast feeding a few hours after birth. He is prescribed a routine intramuscular vitamin K1 injection. Which of the following best explains the reason for this recommendation?

- A. Human milk does not contain any vitamin K
- B. Vitamin K1 injection prevents vitamin K deficiency bleeding (VKDB)
- C. Vitamin K deficiency is common throughout the first year of life
- D. Vitamin K2 but not K1 is found in large amounts in the intestine of newborns
- E. VKDB occurs in half of the newborns and is mostly fatal

4. A 27-year-old woman comes to the physician for a routine post natal examination. She is breast feeding her 1-month-old boy. She says she is eating a diet supplemented with EPA and DHA. This diet increases the content of DHA in human milk which is important for eye development of the infant. Which of the following best explains the significance of this fatty acid to perform this function?

- A. Structural component in rod and cones membranes.
- B. Needed for eicosanoid synthesis.
- C. Needed for the synthesis of arachidonic acid.
- D. Decreases the fluidity in the retinal membranes.
- E. Increases the uptake of vitamin A from the diet.

5. A 6-month-old boy is brought to the physician by his parents for a routine examination. He is breast fed and receives additional solid food. His mother is a vegetarian. She feeds him spinach puree to provide dietary iron. Which of the following conditions most likely increases the dietary uptake and bioavailability of this mineral?

- A. Oxidation of iron to ferric (Fe^{3+}) state
- B. A high pH in the stomach
- C. Phytates found in spinach
- D. Presence of iron in ferrous (Fe^{2+}) state
- E. Prolonged cooking of spinach

6. A 21-year-old woman comes to the physician as she wants to start a family. She is on valproic acid treatment for epilepsy. A side effect of this drug is to reduce the activation of folic acid to tetrahydrofolate. She is advised not to get pregnant while on valproic acid treatment as it could lead to malformations of her fetus. Which of the following could most likely result from abnormal folic acid metabolism due to the anti-epileptic drug?

- A. Blindness
- B. Spina bifida
- C. Congenital Hypothyroidism
- D. Down Syndrome
- E. Cleft lip

7. A 4-day-old girl is brought to the physician for a follow-up examination. She is delivered at full-term in the hospital. Blood and urine samples sent for newborn screening studies showed the presence of galactose in the blood and urine samples. The parents are counseled that their child has classical galactosemia. Which of the following is the most appropriate advice for this patient?

- A. Colostrum can be given but after that formula is needed
- B. Galactose needs to be supplemented
- C. Lactose-free formula is needed instead of human milk
- D. Breast-feeding shall only start after the 6th week
- E. The diet has to be galactose-free but can contain lactose

8. A research group analyzes human milk during their project to develop a specific formula that is most beneficial for the brain development of infants. Additional supplementation with which of the following would most likely improve the development of this organ in infants?

- A. Iron
- B. IgA
- C. Sucrose
- D. Docosahexaenoic acid (DHA)
- E. Calcium

9. A 3-month-old boy is brought to the physician by her mother because she wants to supplement cow's milk in his diet. She is advised to breast feed or use formula as it is easier to digest and not to use cow milk unless it is modified. Which of the following features of breast milk best explains the reason for this advice?

- A. Higher whey/casein ratio
- B. Higher protein amount
- C. Higher IgM/IgA ratio
- D. Lower lactose content
- E. Has twice the amount of fat

10. A 22-year-old woman who has delivered a healthy girl at full-term is advised to breast feed 6 hours after birth. She is advised that the colostrum is very important for the development of her newborn baby. Which of the following best describes this fluid?

- A. Primary function is to provide nutrients for the infants' energy metabolism
- B. IgG represents half of the proteins
- C. Higher in lactose than mature milk
- D. Contains high amount of vitamin A which gives it a yellow color
- E. Produced after first week of lactation

Answer key: 1 C, 2 A, 3 B, 4 A, 5 D, 6 B, 7 C, 8 D, 9 A, 10 D

Short Explanations and comments for practice MCQ

1. A 21-year-old woman comes to the physician because of a 3-week history of missing her period. Laboratory studies confirm that she is pregnant. She says that she is still drinking red wine as she believes that it will not harm her fetus. The physician recommends not to drink any alcohol as this can lead to physical, cognitive and behavioral disabilities in her child. Which of the following is most frequently a complication if she continues to take alcohol?

Comment: The fetus metabolizes ethanol differently than the adult. Ethanol is linked to fatty acids forming fatty acyl ethyl esters which can be measured. Ethanol concentration is high in the placenta and damages the fetus resulting in fetal alcohol syndrome which is characterized by facial dysmorphism

- A. Protruding tongue - **no** this is characteristic feature of congenital hypothyroidism; Also known as macroglossia. It is more common in mothers who have iodine deficiency
- B. Macrosomia - **no** this describes large babies (increased birth weight) and fetal anomalies are complications of gestational diabetes
- C. Facial abnormalities - **YES, correct, midline facial abnormalities are found in fetal alcohol syndrome**
- D. Cleft lip - **no** this is found in when the mother takes in high doses of vitamin A and increased retinoid levels during pregnancy. Note that high doses of vitamin A are teratogenic
- E. Spina bifida - **no** this is found in pre and perinatal folate deficiency in the mother

2. A researcher is studying composition of human milk and observes that it contains the proteins, whey and casein. Together they provide all dietary essential amino acids and in addition **amino acids** that are especially needed in large amounts in infants for **collagen synthesis**. Which of the following amino acids found in human milk is most likely used for this function?

- A. Proline - **Yes Correct Proline is needed in large amounts for collagen synthesis especially in neonates and infants. Proline is not a dietary essential amino acid but it is provided in larger amounts by human milk**

- B. Leucine **no** leucine is a dietary essential amino acid found in human milk but this amino acid is not needed for collagen synthesis. Leucine stimulates muscle growth leading to a positive nitrogen balance
- C. Cysteine **no** cysteine is not found in collagen but is needed for glutathione synthesis. Cysteine is not dietary essential but is needed in large amounts in infants and it is provided by human milk
- D. Tyrosine **no** tyrosine is not found in collagen. It is not an essential amino acid but it is needed in large amounts in infants for hormone and melanin synthesis. Tyrosine is provided in larger amounts by human milk
- E. Histidine **no** Histidine is a dietary essential amino acid but it is not needed for collagen synthesis. Histidine is needed in larger amounts for hemoglobin synthesis and is provided by human milk

3. A boy is delivered at full-term in the hospital. He is healthy and starts breast feeding a few hours after birth. He is prescribed a routine intramuscular **vitamin K1 injection**. Which of the following best explains the reason for this recommendation?

Comment: VKDB is found in some neonates or infants when they are vitamin K deficient. Neonates are prone to vitamin K deficiency as they have a **sterile intestine** and lack bacteria which can provide vitamin K2. **In addition, human milk is low in vitamin K** and food containing vitamin K1 in form of vegetables is mostly given after 6 months of age in addition to milk. Medically, Vitamin K1 is generally injected but can be given orally to neonates. VKDB in newborns is generally not fatal but VKDB can lead to sudden death in some infants at about 6 months of age. Vitamin K is required for the maturation of clotting factors II, VII, IX and X in the liver.

- A. Breast milk does not contain any vitamin K **no** Human milk is low in Vitamin K1 and the vitamin needs to be supplemented
- B. Vitamin K1 injection prevents vitamin K deficiency bleeding (VKDB) **Yes, correct.** Newborns are deficient in Vitamin K as their intestine is sterile and vitamin K2 is not provided by bacteria. In addition, human milk is low in vitamin K1. Injection of vitamin K1 has been shown to protect against VKDB
- C. Vitamin K deficiency is common throughout the first year of life **no** deficiency is found in the neonates but in infants vitamin K is provided by bacteria and diet
- D. Vitamin K2 but not K1 is found in large amounts in the intestine of newborns **no** the intestine of newborns is mainly sterile and later on bacteria (intestinal microbiome) provides vitamin K2
- E. VKDB occurs in half of the newborns and is mostly fatal **no**, VKDB is not that common and in newborns it is mostly not fatal. About 3% of newborns present with severe bleeding and it may be fatal in some infants. Vit K supplements immediately after birth can prevent the occurrence of this severe complication.

4. A 27-year-old woman comes to the physician for a routine post natal examination. She is breast feeding her 1-month-old boy. She says she is eating a diet supplemented with EPA and DHA. She has read that such a diet increases the content of DHA in breast milk which is important for eye development in the infant. Which of the following best explains the significance of this fatty acid to perform this function?

- A. Structural component in rod and cones membranes **Yes, Correct** DHA 22:6 ω 3 provides the necessary fluidity and structure of the membranes
- B. Needed for eicosanoid synthesis **no** Eicosanoids are synthesized from a fatty acid with 20 carbons. DHA has 22 carbons and is not used
- C. Needed for the synthesis of arachidonic acid **no** Arachidonic acid is needed the membranes but it **cannot** be formed from DHA. Arachidonic acid is an omega-6 fatty acid (20:4) and can be formed from linoleic acid
- D. Decreases the fluidity in the retinal membranes **no** It **increases** the fluidity
- E. Increases the uptake of vitamin A from the diet **no**. Vitamin A is important for vision but it does not need DHA to increase its uptake from the diet.

5. A 6-month-old boy is brought to the physician by his parents for a routine examination. He is breast fed and receives additional solid food. His mother is a **vegetarian**. She feeds him spinach puree to provide dietary iron. Which of the following conditions most likely increases the dietary uptake and bioavailability of this mineral?

Comment: The uptake of inorganic iron is more difficult than the uptake of heme iron which can be in the ferrous or also in the ferric state (then heme is named hemin). Inorganic iron has to be in the ferrous state for uptake into the intestinal mucosal cells. Low pH in stomach, dietary vitamin C facilitate iron absorption by maintaining iron in the ferrous state. Iron metabolism is very complex and will be taught in more detail in the DM module.

- A. Oxidation of iron to ferric (Fe^{3+}) state **no** it has to be Fe^{2+} (ferrous instead of ferric form). The ferrous form of iron is better absorbed than the ferric form.
- B. A high pH in the stomach **no** Effective iron absorption needs a low pH with high proton concentration. Low pH facilitates the reduction the iron to the ferrous state
- C. Phytates found in spinach **no** the uptake of inorganic iron is decreased by phytates
- D. Presence of iron in ferrous (Fe^{2+})state **Yes, correct.** Dietary iron is changed to the ferrous state in the stomach for later uptake (absorption) into the intestinal mucosal cells
- E. Prolonged cooking of spinach **no** the main factor is that the inorganic iron has to be in the ferrous state for its absorption

6. A 21-year-old woman comes to the physician as she wants to start a family. She is on **valproic acid** treatment for epilepsy. A side effect of this drug is to reduce the activation of folic acid to tetrahydrofolate. She is advised not to get pregnant while on valproic acid treatment as it could lead to malformations of her fetus. Which of the following could most likely result from abnormal folic acid metabolism due to the anti-epileptic drug?

Comment: Pre and perinatal deficiency of folic acid and tetrahydrofolate can lead to spina bifida in the fetus. This is especially important in the first trimester. Zinc deficiency has a similar outcome.

- A. Blindness **no** this can result from toxoplasmosis
- B. Spina bifida **YES, correct. Especially in the first trimester, folate deficiency and specific drugs that interfere with folate metabolism can lead to spina bifida**
- C. Congenital Hypothyroidism **no** this results from iodine deficiency during pregnancy
- D. Down Syndrome **no** The risk of trisomy 21 (fetal aneuploidy) increases with an increase in maternal age due to higher risk of non-disjunction during meiosis (oogenesis)
- E. Cleft lip **no** this can result from extremely high vitamin A and retinoid levels during pregnancy

7. A 4-day-old girl is brought to the physician for a follow-up examination. She is delivered at full-term in the hospital. Blood and urine samples sent for newborn screening studies showed the presence of galactose in the blood and urine samples. The parents are counseled that their child has classical galactosemia. Which of the following is the most appropriate advice for this patient?

Comment: Classic Galactosemia is a severe hereditary disease that can lead to liver, brain and kidney damage, and to cataracts. Nutrition without galactose is needed, for the infant, lactose is main nutrient from milk which has to be substituted by a specific lactose and galactose free formula. It is critical to avoid lactose or galactose as early as possible to prevent any developmental delay.

- A. Colostrum can be given but after that formula is needed **no colostrum has lactose, less than mature milk but still should not be given**
- B. Galactose needs to be supplemented **no galactose has to be excluded from the diet**
- C. Lactose-free formula is needed instead of human milk **YES, correct, the baby needs a lactose-free formula instead of human milk**
- D. Breast-feeding shall only start after the 6th week **no colostrum, transition and mature milk all contain lactose**

- E. The diet has to be galactose-free but can contain lactose **no** both lactose and galactose have to be avoided in the diet

8. A research group analyzes human milk during their project to develop a specific formula that is most beneficial for the brain development of infants. Additional supplementation with which of the following would most likely improve the development of this organ in infants?

Comment: Different formulae are now on the market which use cow milk which is then modified and supplemented to be similar to human milk.

- A. Iron **no** it is added in some formulae; but is low in human milk to deprive intestinal microbes of iron
- B. IgA **no** it is high in human milk but does not directly improve brain development. It protects the lining of the intestines and prevents infection in infants
- C. Sucrose **no** it is added to some formulas but it does not improve brain development, it helps energy metabolism
- D. Docosahexaenoic acid (DHA) **YES, correct, brain development continues after birth and DHA improves it. DHA is an omega-3 fatty acid**
- E. Calcium **no** calcium is mainly needed for teeth and bone development

9. A 3-month-old boy is brought to the physician by her mother because she wants to supplement cow's milk in his diet. She is advised to breast feed or use formula as it is easier to digest and not to use cow milk unless it is modified. Which of the following features of breast milk best explains the reason for this advice?

Comment: It is mainly the high casein/whey ratio in cow milk that makes it difficult to digest by infants. Some formula have added whey protein. If absolutely needed, cow milk can be used in diluted form. The question is testing the differences between cow's milk and breast milk.

- A. Higher whey/casein ratio **YES, correct. Whey is easier to digest, whereas casein leads to curd**
- B. Higher protein amount **no** it has a lower protein content
- C. Higher IgM/IgA ratio **no** it has mainly IgA
- D. Lower lactose content **no** it has a higher lactose content
- E. Has twice the amount of fat **no** The fat content is similar in breast and cow's milk

10. A 22-year-old woman who has delivered a healthy girl at full-term is advised to breast feed 6 hours after birth. She is advised that the colostrum is very important for the development of her newborn baby. Which of the following best describes this fluid?

Comment: Colostrum (first milk) protects the infant against infections and supports growth. Mature milk then more provides nutrients.

- A. Primary function is to provide nutrients for the infants' energy metabolism **no**
primary function is to support the immune system and facilitate growth
- B. IgG represents half of the proteins **no** it is rich in IgA
- C. Higher in lactose than mature milk **no** lower in lactose than mature milk, primary function is not provision of macro nutrients but IgA and lactoferrin
- D. Contains high amount of vitamin A which gives it a yellow color **YES, correct.**
Retinoids formed from vitamin A are needed for many functions, including cell division and growth
- E. Produced after first week of lactation **no** produced during pregnancy and up to 6 days postpartum

Practice questions: Nutritional disorders in adolescence and childhood

1. A 3-year-old boy is brought to the physician by his mother for a follow-up examination. Three months ago, he was diagnosed with Retinoblastoma (cancer of the retina). Several cycles of chemotherapy have been given which were fairly tolerated. He is underweight, malnourished, anorexic and lethargic. Current weight is less than 60% of the expected for age, height has remained unchanged since the onset of the current illness, skin is intact. Which of the following best describes the patient?
 - A. Kwashiorkor
 - B. Marasmus
 - C. Cachexia
 - D. Marasmic-kwashiorkor
 - E. Stunting
2. A 24-month-old girl is brought to the physician by her parents because of a 2-week history of swelling of the hands and feet, poor appetite and irritability. She lives in rural sub-Saharan Africa. She was weaned off breast milk 2 months earlier. Dietary history shows a predominantly maize-based diet. Her vital signs are within normal limits. Physical examination shows bilateral

pitting pedal edema. With sustained, firm pressure, the imprint of the examiner's thumb is seen and an 8-mm depression is created. She has a dry, flaky rash on the lower extremities. Urinalysis is negative for protein. Which of the following tests will be most appropriate for assessing long-term malnutrition in this patient?

- A. Serum Retinol binding protein
- B. Serum Prealbumin
- C. Serum Transferrin
- D. Serum Albumin
- E. Serum Folate levels

3. An 18-month-old boy is brought to the physician by his mother for a follow-up examination. He was treated 2 weeks ago for a recurrent chest infection. He was diagnosed with Cystic fibrosis at the age of 6 months and has had several hospital admissions. Anthropometric parameters show a failure to thrive. Physical examination shows anorexia, lethargy, with a fever of 102°F, dry skin, and buccal mucosa. Which of the following terms best describes the weight category of this patient?

- A. Cachexia
- B. Kwashiorkor
- C. Marasmus
- D. Marasmic-kwashiorkor
- E. Stunting

4. A 3-year-old boy is brought to the physician by his parents because of a 5-day history of worsening swelling of hands and feet. He had a measles infection 2 weeks ago which was complicated by diarrhea but no vomiting. He weighs about 75% of the expected weight for age. Physical examination shows sore, reddish eyes and photophobia. Physical examination shows desquamating dermatitis, particularly on the lower legs, where there is pitting edema. He has 3 cm hepatomegaly and is apathetic. Which of the following is the most likely diagnosis?

- A. Marasmus
- B. Kwashiorkor
- C. Marasmic-kwashiorkor
- D. Cachexia
- E. Stunting

5. A 2-year-old boy is admitted to the hospital for management of severe malnutrition. He develops signs of heart failure within 24-hour of the commencement of dietary rehabilitation. Laboratory studies show hypophosphatemia, hypokalemia, and hypomagnesemia. He is transferred to the intensive care unit for subsequent management. Which of the following approaches would most likely have prevented this complication?

- A. Correct macro-nutrient abnormalities early

- B. Treat infections first
- C. Correct electrolyte abnormalities first
- D. Withhold treatment till after 48hrs
- E. Give adequate protein and carbohydrates

Answer key:

- 1. C
- 2. D
- 3. A
- 4. B
- 5. C

Short explanations

1. A 3-year-old boy is brought to the physician by his mother for a follow-up examination. Three months ago, he was diagnosed with Retinoblastoma (cancer of the retina). Several cycles of chemotherapy have been given which were fairly tolerated. He is underweight, malnourished, anorexic and lethargic. Current weight is less than 60% of the expected for age, height has remained unchanged since the onset of the current illness, skin is intact. Which of the following best describes the patient?

Comment: Cachexia can be seen in patients with chronic illnesses, cancers, infections where weight loss in this setting cannot be easily reversed by dietary rehabilitation alone. A, B, D are distractions which can occur in similar settings (refugee camp, malnourished, underweight). However, with an underlying diagnosis of a cancer cachexia is more apparent.

- A. Kwashiorkor – Skin lesions and hair changes are usually present; Patients have edema
- B. Marasmus – Severe weight loss; extensive loss of adipose tissue; No edema
- C. Cachexia – Correct answer – Presence of the inflammatory mediators increases the risk of malnutrition**
- D. Marasmic-kwashiorkor – Have some features of marasmus and some of kwashiorkor; pedal edema is present; Skin lesions and hair changes may or may not be present
- E. Stunting – Less height of the patient compared to controls; no skin lesions; no edema

2. A 24-month-old girl is brought to the physician by her parents because of a 2-week history of swelling of the hands and feet, poor appetite and irritability. She lives in rural sub-Saharan Africa. She was weaned off breast milk 2 months earlier. Dietary history shows a predominantly maize-based diet. Her vital signs are within normal limits. Physical examination shows bilateral pitting pedal edema. With sustained, firm pressure, the imprint of the examiner's thumb is seen and an 8-mm depression is created. She has a dry, flaky rash on the lower extremities. Urinalysis is negative for protein. Which of the following tests will be most appropriate for assessing long-term malnutrition in this patient?

Comment: The patient has Kwashiorkor. Note the pitting edema and skin lesions. Severe protein (essential amino acid) deficiency usually results in kwashiorkor.

- A. Serum Retinol binding protein – may be used as short-term markers
- B. Serum Prealbumin - may be used as short-term markers
- C. Serum Transferrin
- D. Serum Albumin – Correct answer - serum albumin is the most reliable of all the given options to serve as a marker for long-term malnutrition with a half-life of approximately 20 days.**
- E. Serum Folate levels

3. An 18-month-old boy is brought to the physician by his mother for a follow-up examination. He was treated 2 weeks ago for a recurrent chest infection. He was diagnosed with Cystic fibrosis at the age of 6 months and has had several hospital admissions. Anthropometric parameters show a failure to thrive. Physical examination shows anorexia, lethargy, with a fever of 102°F, dry skin, and buccal mucosa. Which of the following terms best describes the weight category of this patient?

- A. Cachexia – Correct answer – Complications of Cystic fibrosis is the cause of the malnutrition which is complicated by the recurrent infections**
- B. Kwashiorkor
- C. Marasmus
- D. Marasmic-kwashiorkor
- E. Stunting

4. A 3-year-old boy is brought to the physician by his parents because of a 5-day history of worsening swelling of hands and feet. He had a measles infection 2 weeks ago which was complicated by diarrhea but no vomiting. He weighs about 75% of the expected weight for age. Physical examination shows sore, reddish eyes and photophobia. Physical examination shows desquamating dermatitis, particularly on the lower legs, where there is pitting edema. He has 3 cm hepatomegaly and is apathetic. Which of the following is the most likely diagnosis?

Comment: Although there is an underlying illness (measles), it is an acute one affecting the current utilization of nutrients. Complete recovery is possible and nutritional supplementation is very vital for recovery.

- A. Marasmus – Children do not have pedal edema; Extensive loss of adipose tissue in face and limbs
- B. Kwashiorkor – Correct answer – Note the pedal edema, skin lesions, hepatomegaly which indicate kwashiorkor.**
- C. Marasmic-kwashiorkor

- D. Cachexia
- E. Stunting

5. A 2-year-old boy is admitted to the hospital for management of severe malnutrition. He develops signs of heart failure within 24-hour of the commencement of dietary rehabilitation. Laboratory studies show hypophosphatemia, hypokalemia, and hypomagnesemia. He is transferred to the intensive care unit for subsequent management. Which of the following approaches would most likely have prevented this complication?

Comment: Refeeding syndrome is a complication of overzealous correction of malnutrition. A stepwise approach is always the best option, starting with electrolyte correction, treatment of infections if any before providing macronutrients.

- A. Correct macro-nutrient abnormalities early
- B. Treat infections first
- C. **Correct electrolyte abnormalities first – Correct answer**
- D. Withhold treatment till after 48hrs
- E. Give adequate protein and carbohydrates